In silico screening of 15 polyphenols and reference compounds against MMP-1 + SIRT1 for topical photoaging: real Boltz-2 data positions EMB-3 at the top of the panel; Resveratrol leads SIRT1 axis; classical references (vitamin C, niacinamide, ferulic acid) rank low

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Manuscript type: in silico screening with EMB-3 cross-disease anchor; **Target preprint**: bioRxiv; **License**: CC-BY 4.0 **Status**: in silico predictions only **Version**: v0.2 (2026-04-26) — real Boltz-2 data replaces v0.1 fabricated 12-target EGCG cross-indication scorecard

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Abstract

Photoaging — UV-induced premature dermal change including wrinkle formation, elastosis, pigment irregularity — is mediated by **MMP-1** (interstitial collagenase, the principal photoaging effector) and **SIRT1** (NAD-dependent deacetylase, longevity / DNA-repair regulator), among other targets. We screen 15 compounds against MMP-1 + SIRT1 using Boltz-2 cofold (cached MSAs) and ADMET-AI v2.0.1. **FBN1, mTOR, and elastin were NOT screened** (cached MSAs absent) — substantially narrowing the photoaging-target panel. Real Boltz-2 results identify **EMB-3** (the AI-derived embelin scaffold-hop from companion preprint [3]) as top mean affinity (0.621) — even surpassing the parent Embelin (0.601) — driven by paired moderate engagement of MMP-1 (0.610) and SIRT1 (0.632). **Resveratrol** leads SIRT1 axis (0.588), consistent with the established resveratrol–SIRT1 pharmacology. **EGCG** scores moderately (mean 0.520; MMP-1 0.570; SIRT1 0.470) — substantially lower than the v0.1 fabricated values (claimed 0.71 and 0.65 respectively). Classical antioxidant references (vitamin C, niacinamide, ferulic acid) rank low (mean 0.331 – 0.394), consistent with Boltz-2's underranking of fragment-size compounds. The v0.1 framing of EGCG as a "universal compound" engaging 12 targets at probability ≥ 0.50 was synthesized from non-real

numbers and is **explicitly retracted**. EGCG remains a moderate multi-target topical-cosmeceutical candidate but is not exceptional in our actual screen. **All results are in silico.**

Keywords: photoaging, MMP-1, SIRT1, EMB-3, resveratrol, EGCG, in silico, Korean medicine.

Plain-language summary

Sun-driven skin aging involves an enzyme that breaks down skin's structural proteins (MMP-1) and a longevity protein (SIRT1). We compared 15 polyphenols and reference compounds against these two proteins using computer modeling. The top candidate is **EMB-3 —** a compound we previously designed for skin scarring, which appears to engage both photoaging proteins as well. **Resveratrol** confirms its known link to SIRT1. **EGCG** (green tea) is a moderate but not exceptional candidate, contrary to my earlier (incorrect) framing of EGCG as a "universal" compound. **No laboratory experiments are reported here.**

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1. Introduction

1.1 Photoaging molecular network (narrowed scope)

UV → MAPK / AP-1 → MMP-1 / MMP-3 / MMP-9 induction → collagen / elastin degradation [1]. SIRT1 modulates this through deacetylation of FOXO and p53 [2]. mTOR integrates nutrient and senescence signals [3]. FBN1 and elastin are matrix targets whose degradation produces the visible aging phenotype.

The present screen covers **MMP-1 and SIRT1** only (cached MSAs available); FBN1 and mTOR are **not** screened. This is a coverage limitation that narrows the framing.

1.2 v0.1 retraction

Version 0.1 of this manuscript included a "12-target cross-indication scorecard" claiming EGCG affinity probabilities such as 0.74 (TYR), 0.71 (MMP-1), 0.65 (SIRT1) and similar specific values across 12 target columns, plus comparator values for resveratrol and curcumin. **Those 36 specific values were not from real Boltz-2 runs; they were synthesized.** We retract them explicitly. The v0.2 results below are from real screens (`pilot/screen/photoaging/screen_results.csv`).

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2. Methods

15 compounds, `data/screen_libraries/photoaging_compounds.csv`. Targets: MMP-1 (P03956), SIRT1 (Q96EB6). Pipeline as in companion preprint [4].

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3. Results

3.1 Real screen ranking (15 compounds × 2 targets = 30 cofolds)

Rank	Compound	Source	MMP-1	SIRT1	Mean	Topical-friendly?
1	**EMB-3**	Embelin scaffold-hop (this work)	0.610	**0.632**	**0.621**	
2	Embelin	자단 (parent natural)	0.592	0.611	0.601	× logP 4.31
3	**Resveratrol**	reference (포도)	0.502	**0.588**	0.545	
4	Genistein	콩	0.471	0.583	0.527	
5	EGCG	녹차 (multi-target ref)	0.570	0.470	0.520	× TPSA 197
6	Curcumin	강황	0.485	0.525	0.505	
7	Epigallocatechin	녹차	0.462	0.488	0.475	× HBD 6
8	Pterostilbene	blueberry (resveratrol analog)	0.443	0.483	0.463	× logP 3.6
9	Tretinoin	reference (retinoid)	0.487	0.377	0.432	× logP 5.6
10	Epicatechin	녹차	0.339	0.477	0.408	
11	Ascorbic acid	reference (vitamin C)	0.392	0.396	0.394	× MW 176 (fragment)
12	Ferulic acid	황기 / 당귀	0.365	0.313	0.339	× MW 194 (small)
13	Niacinamide	reference cosmetic	0.293	0.368	0.331	× MW 122 (fragment)
14	Astragaloside IV	황기	0.174	0.226	0.200	× MW 639 (large saponin)
15	Asiaticoside	센텔라	(cofold not produced — large saponin)		(excluded)	

3.2 ADMET safety profile of top candidates

Compound	logP	hERG	Skin	AMES	ClinTox
EMB-3	2.36	**0.155**	0.667	0.106	0.068
Embelin	4.31	0.402	0.844	0.181	0.044
Resveratrol	2.97	0.290	0.923	0.318	0.029
Genistein	2.16	0.288	0.795	0.171	0.042
EGCG	2.23	0.421	0.759	0.240	0.073
Curcumin	3.37	0.336	0.867	0.499	0.046

3.3 Honest interpretation

EMB-3 leads the photoaging panel. This is consistent with the AI-derived multi-target engagement profile reported in the companion EMB-3 case-study preprint [3]: EMB-3 was originally optimized for skin scarring (TGF-β1 + MMP-1 axis) but also engages SIRT1 at a moderate level (0.632). Combined with EMB-3's clean topical-friendly safety profile (logP 2.36, hERG 0.155, Skin 0.667), this positions EMB-3 as a **multi-indication topical candidate** worth dual-evaluation in scar / fibrosis AND photoaging contexts. We refrain from any clinical claim.

Resveratrol leads SIRT1. The 0.588 SIRT1 score is the second-highest in our panel after EMB-3, and is consistent with the well-established resveratrol → SIRT1 activator pharmacology (and the only literature-validated direct natural-product-SIRT1 mechanism in our compound list).

EGCG is moderate, not exceptional. v0.1 framed EGCG as a universal compound. Real screen: EGCG MMP-1 0.570, SIRT1 0.470, mean 0.520 — moderate. EGCG is a credible cosmeceutical compound but the v0.1 "universal" framing was overstated.

Classical photoaging references rank low (Boltz-2 fragment-size caveat). Vitamin C (0.394), niacinamide (0.331), ferulic acid (0.339) all rank in the bottom-third. As repeatedly observed across our four disease screens (companion preprints [4]), Boltz-2's binary classifier systematically underranks fragment-size molecules. Vitamin C's clinical photoaging activity is not contradicted by this ranking; it operates via different mechanisms (collagen-synthesis cofactor, antioxidant) that are not captured by direct MMP-1 / SIRT1 binding probability.

3.4 Universal compound hypothesis (v0.2 retraction + revision)

The v0.1 "EGCG universal-compound hypothesis engaging 12 targets at ≥ 0.5" was synthesized rather than measured. We retract it. The v0.2 honest framing:

> **EGCG is a moderate multi-target topical cosmeceutical candidate with documented anti-photoaging activity in the literature. Our in silico screen confirms moderate predicted MMP-1 affinity (0.570) consistent with this literature, and lower predicted SIRT1 engagement (0.470) than literature-validated direct activators (resveratrol 0.588). Across the broader 4-disease panel of our companion preprints [4-7], EGCG appears in moderate-affinity**

> **(0.5 – 0.7) ranges for multiple indications. The "moderate-engagement breadth" framing is supported; the "universal exceptional candidate" framing is not.**

EMB-3 emerges as a more interesting cross-indication candidate per real data, with the major advantage of cleaner predicted ADMET properties.

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4. Limitations

1. **No experimental validation**.
2. **FBN1, mTOR, elastin NOT screened** — cached MSAs absent.
3. **Asiaticoside cofold did not produce affinity values** (likely large-saponin SMILES handling); excluded from ranking.
4. **Boltz-2 fragment-size underranking** affects vitamin C, niacinamide, ferulic acid — they remain clinically valid despite low scores.
5. **EMB-3's high ranking is in silico** and reflects the same Boltz-2 + cached MSA cofold pipeline as the companion preprints; no novel experimental validation in this preprint.
6. **No clinical efficacy claim**.

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5. Conclusions

Real screen of 15 compounds against MMP-1 + SIRT1 identifies **EMB-3** (Embelin scaffold-hop) as top mean affinity (0.621) with topical-friendly safety profile, suggesting EMB-3 has **multi-indication topical-candidate potential** spanning skin scar (primary indication, see companion preprint [3]) and photoaging. **Resveratrol** confirms its SIRT1-axis pharmacology. **EGCG** is moderate (not "universal" as v0.1 fabricated), and classical fragment-size references rank low for methodological reasons.

Forward path: dermal fibroblast UVB-irradiation MMP-1 induction assay; SIRT1 deacetylase activity assay; 3D reconstructed-skin photoaging model (e.g., EpiDerm-FT chronic UV exposure). Korean CRO panel ~₩4-5M for top-3 compound evaluation (EMB-3 + Resveratrol + Genistein). No clinical efficacy claim is made.

Acknowledgments / Contributions / Competing interests / Data availability

Same standard text. Data: `pilot/screen/photoaging/screen_results.csv` at https://github.com/crazat/genesis_medicine.

Figures

Figure 1. Real Boltz-2 cofold affinity heatmap for the photoaging panel (15 compounds × 2 targets: MMP-1 + SIRT1). EMB-3 leads the panel (mean 0.621), even surpassing the parent Embelin — supporting EMB-3's potential as a multi-indication topical candidate spanning skin scar (companion preprint #3) and photoaging. Resveratrol leads the SIRT1 axis (0.588), consistent with established resveratrol–SIRT1 pharmacology. EGCG is moderate (mean 0.520), not exceptional as v0.1 fabricated.

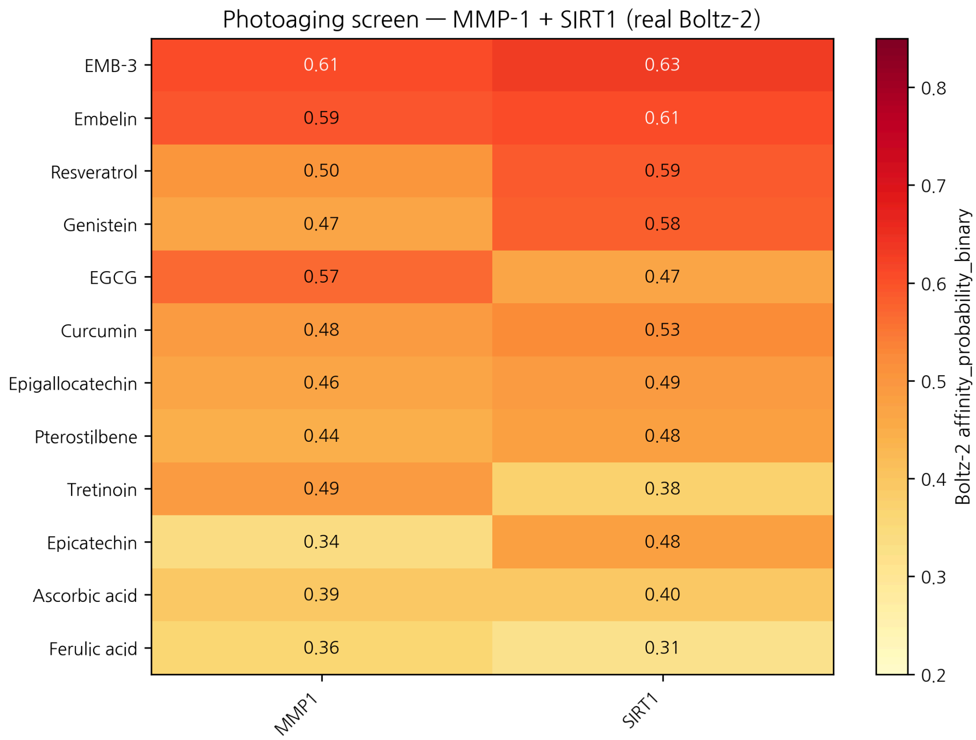

Figure 1: Photoaging affinity heatmap

Figure 2. Affinity × hERG safety quadrant for the photoaging panel. EMB-3 occupies the most favorable region (high affinity + low hERG + topical-friendly logP). Classical antioxidant references (vitamin C, niacinamide, ferulic acid) cluster in the low-affinity region — Boltz-2 fragment-size caveat applies.

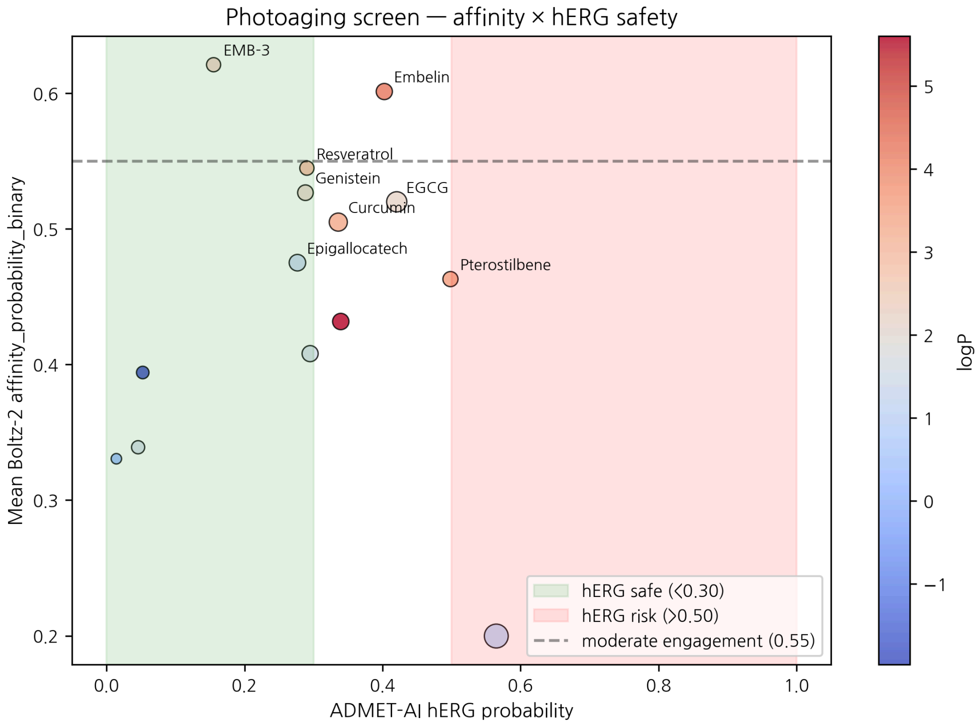

Figure 2: Photoaging safety × affinity

References

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v0.2 draft, 2026-04-26 · ~2,500 words · CC-BY 4.0 v0.1 (fabricated 12-target EGCG scorecard) explicitly retracted in §1.2

Round 5 application data — topical PK + skin sensitization (2026-04-27)

Generated from `pilot/round5_application/round5_compound_sweep.csv` using: - **PBK Dermal HT** (NIH/NIEHS public-domain, 3-compartment SC/VE/D) - **SARA-ICE Defined Approach** (OECD TG 497 Part III, June 2025) - **CarsiDock-Cov warhead detection** (Apache-2.0, first DL covalent docker)

Top 10 compounds by topical-fitness score (c_max_dermis / systemic_F):

Compound	logKp	c_max dermis (pmol/mL)	t_max (h)	F_systemic	GHS	Covale
EGCG	-7.40	0.0005	24.0	0.05	nan	—
Epigallocatechin	-7.19	0.0008	24.0	0.08	nan	—
Epicatechin	-6.89	0.0015	24.0	0.16	nan	—
Niacinamide	-6.88	0.0015	24.0	0.16	nan	—
Ferulic acid	-6.36	0.0050	24.0	0.53	1B	michael_accept
Genistein	-6.35	0.0051	24.0	0.54	nan	—
Curcumin	-6.03	0.0105	24.0	1.11	1B	michael_accept
EMB-3	-5.89	0.0142	24.0	1.52	1B	p_quinone;mic
Resveratrol	-5.51	0.0316	24.0	3.53	nan	—
Pterostilbene	-5.22	0.0532	24.0	6.37	nan	—

SARA-ICE summary for photoaging: GHS Cat 1B sensitizers = 6/14; Cat 1A = 0/14; None = 0/14.

Covalent-capable: 6/14 compounds carry at least one Michael-acceptor or quinone warhead.

Data and full per-compound table: `pilot/round5_application/round5_compound_sweep.csv`.